

Rotor and Wake Aerodynamics

Lifting line model and its application to
a rotor and wake simulation



Learning objectives

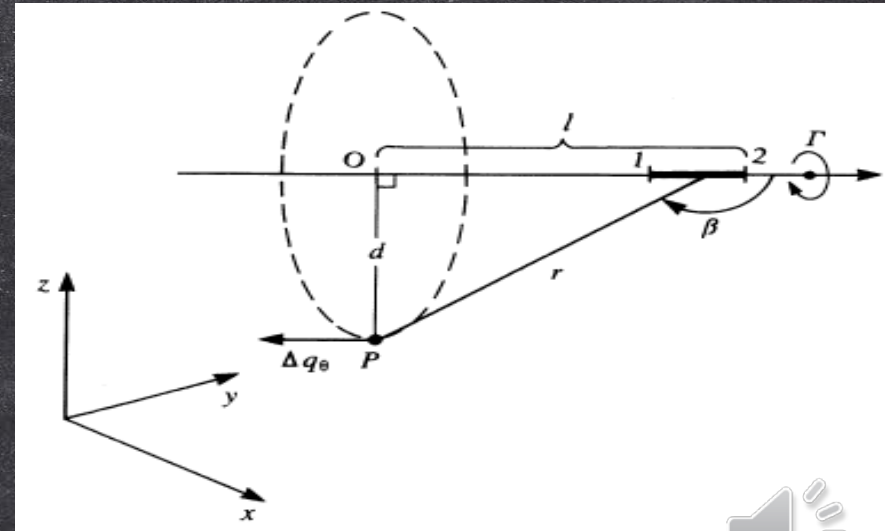
- Identify the main elements of a lifting line model
- Define the boundary conditions necessary to setup the model
- Discuss options to simplify the lifting line model



Vortex Line/Filament Methods

In line vortex or vortex filaments, vorticity is concentrated along material lines of constant circulation

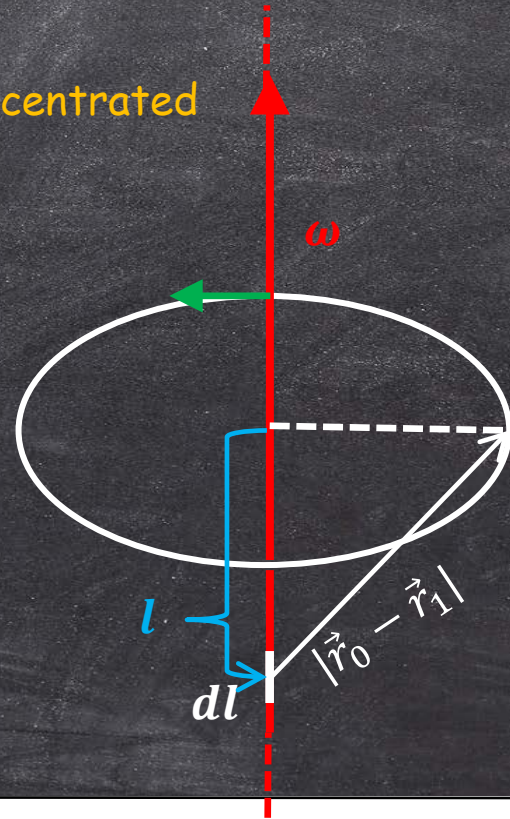
$$\vec{v}_{(\vec{x}_0)} = \Gamma \int_0^l \frac{d\vec{l}_{(\vec{x})} \times (\vec{r}_0 - \vec{r}_1)}{4\pi |\vec{r}_0 - \vec{r}_1|^3}$$



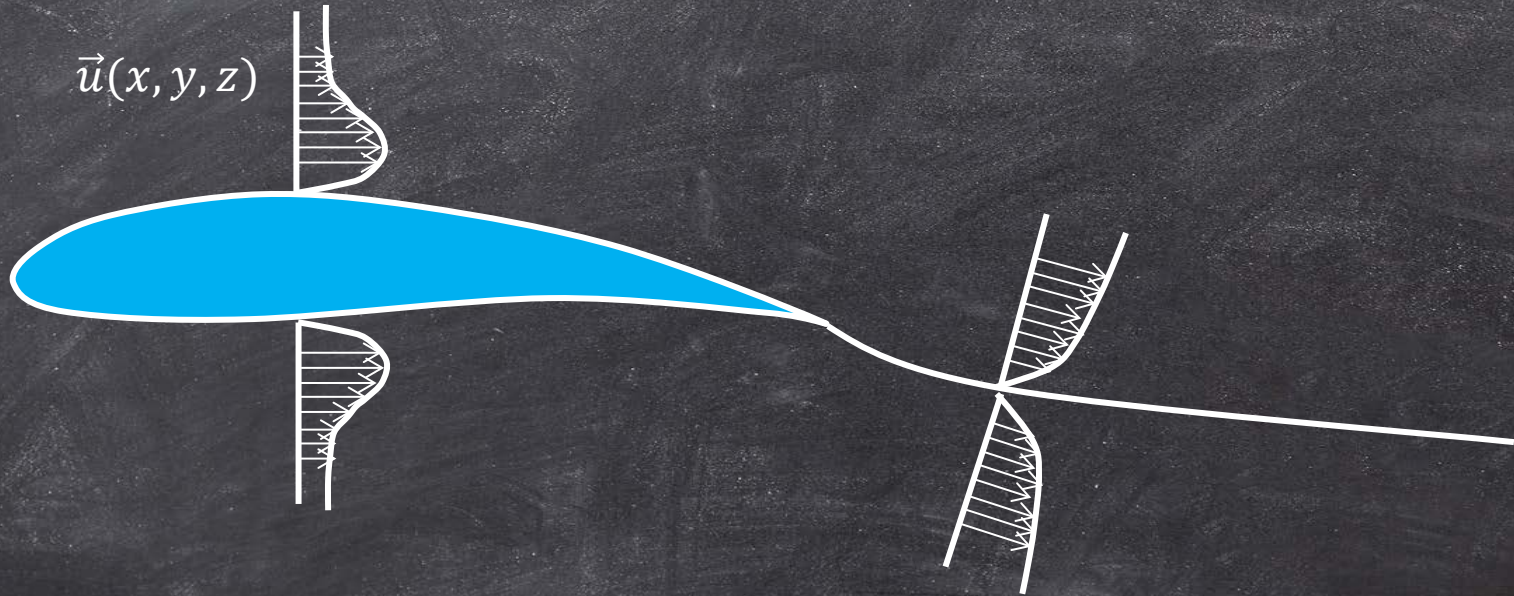
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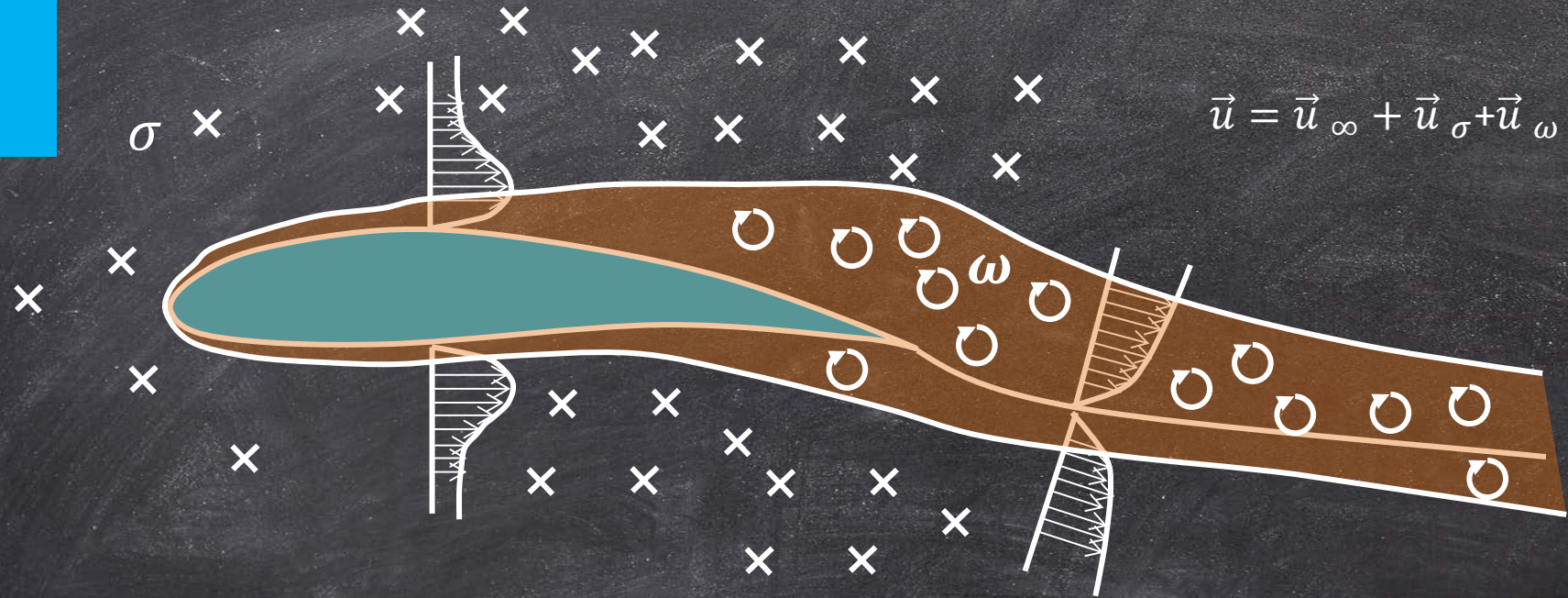
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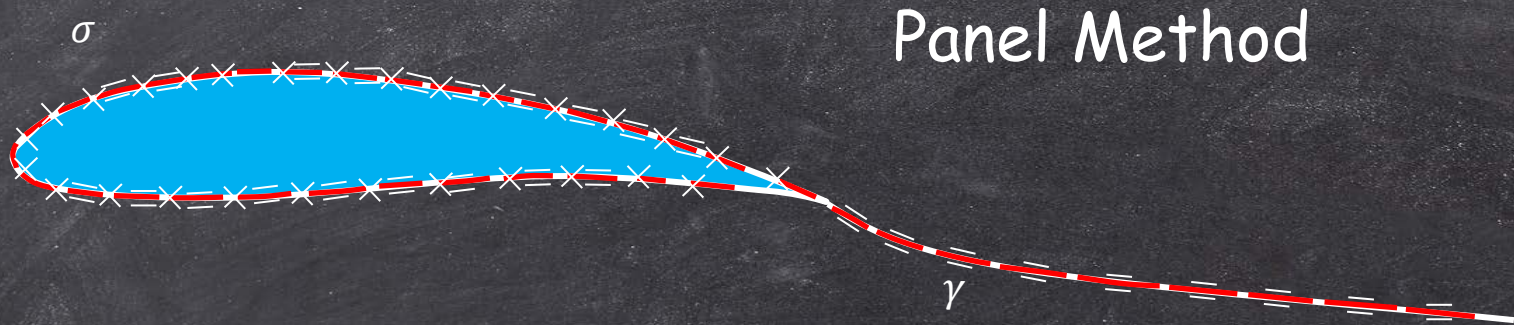
Flow Field Approximations: Panel Method, Lifting Surface, Lifting Line



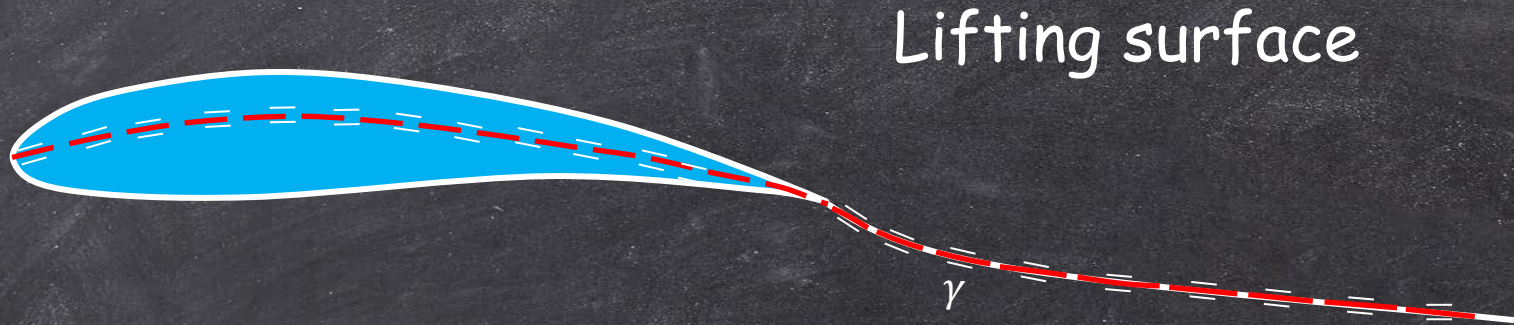
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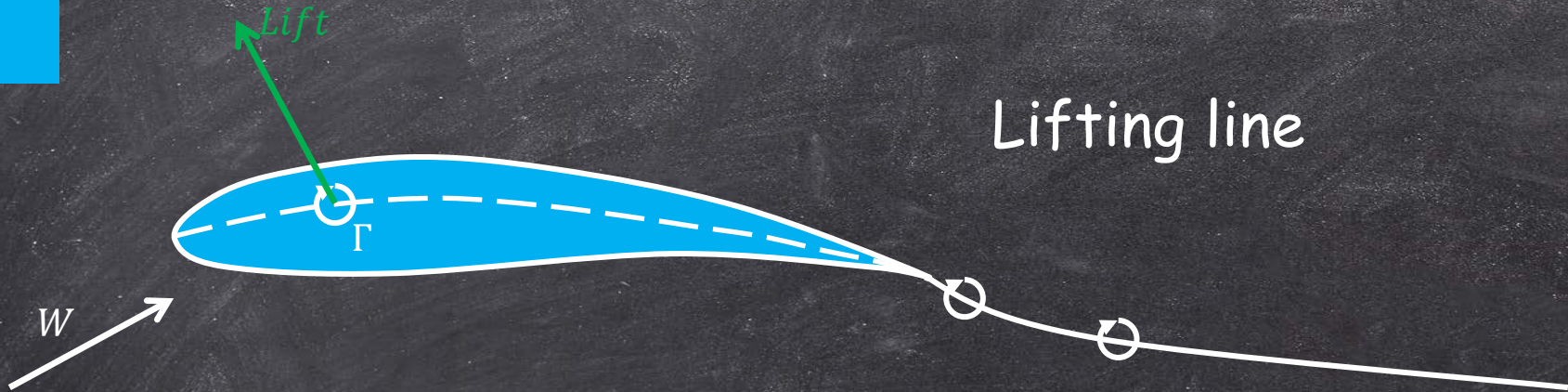
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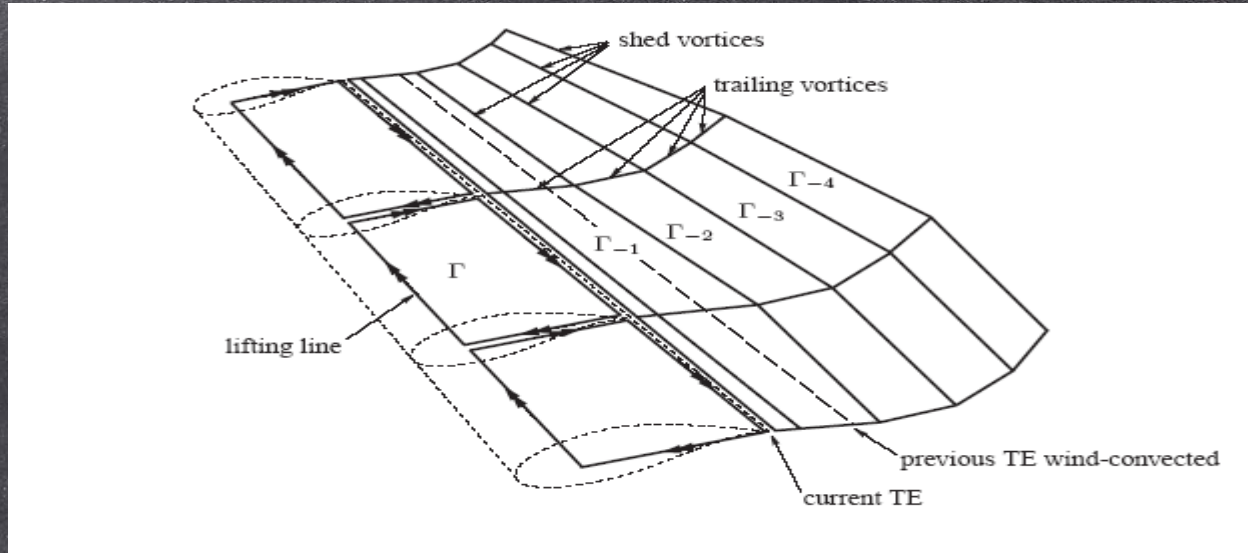
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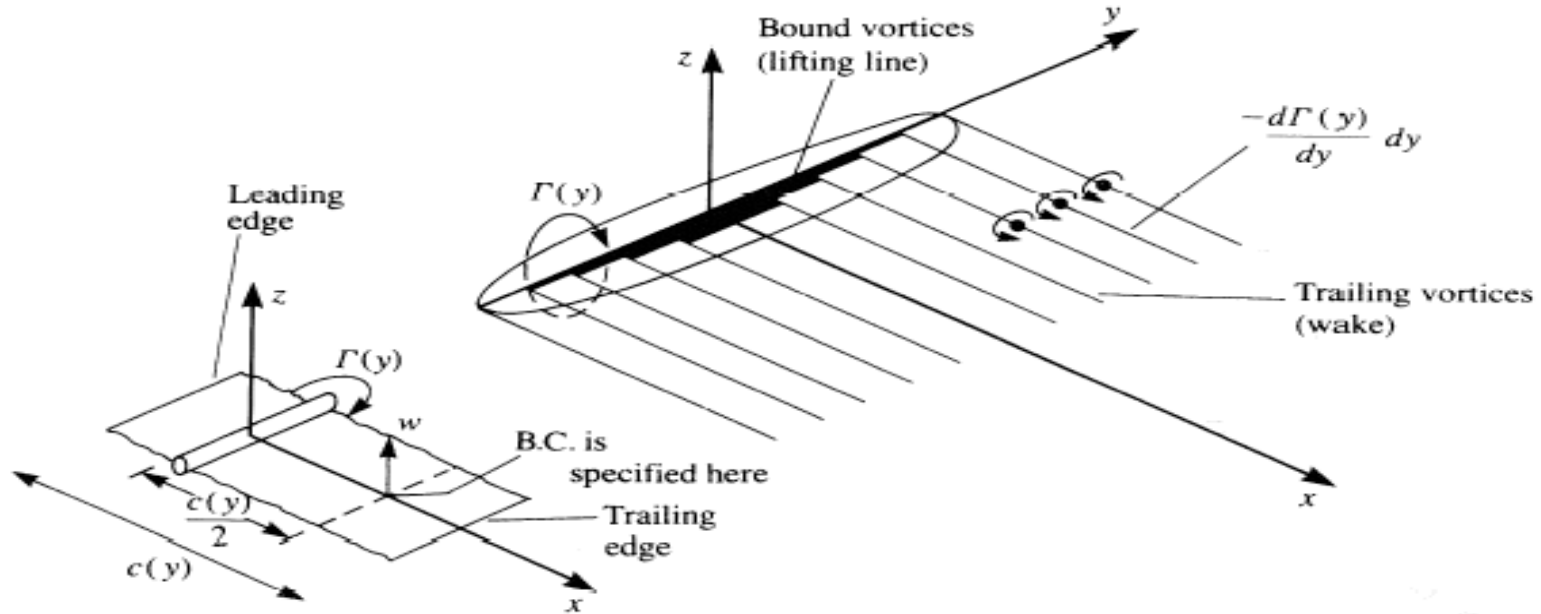
The vortex wake structure is composed of trailing and shed vortices



A. Van Garrel, "Development Of A Wind Turbine Aerodynamics Simulation Module", ECN-C--03-079, 2003



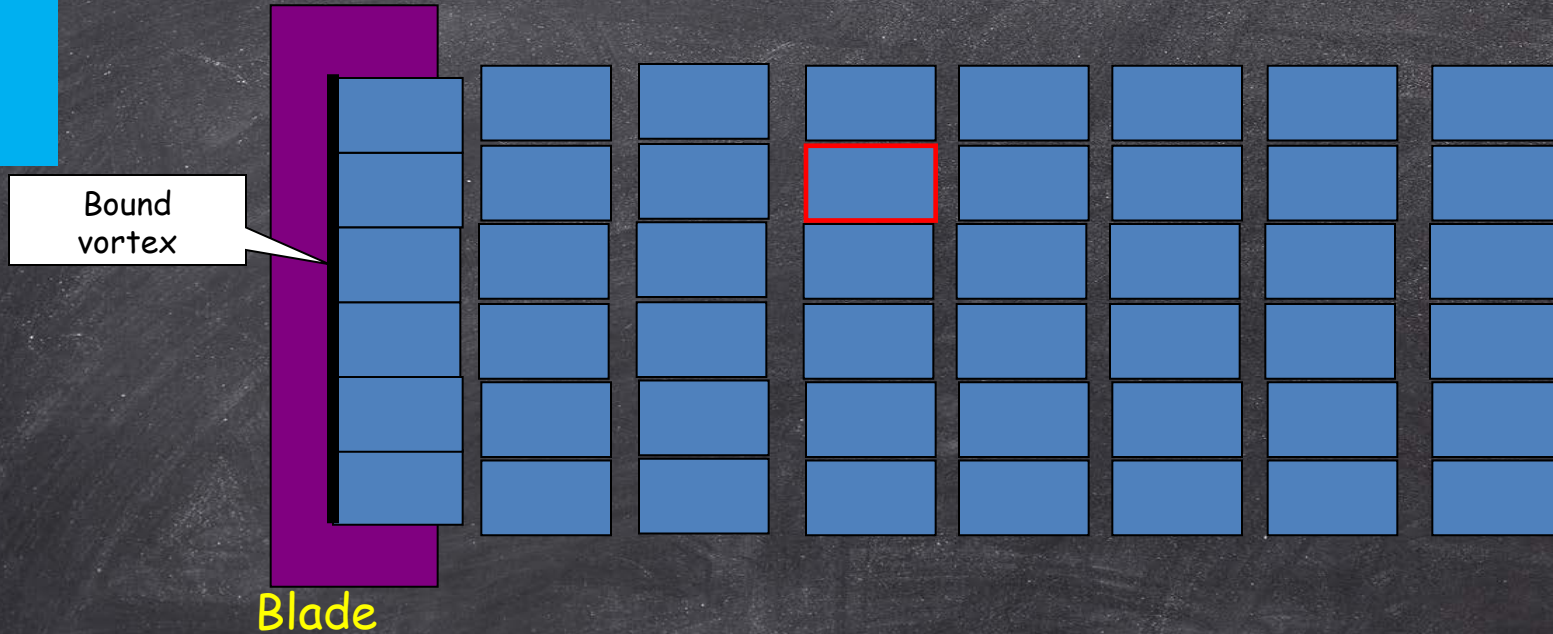
Horseshoe vortex method



Vortex wake



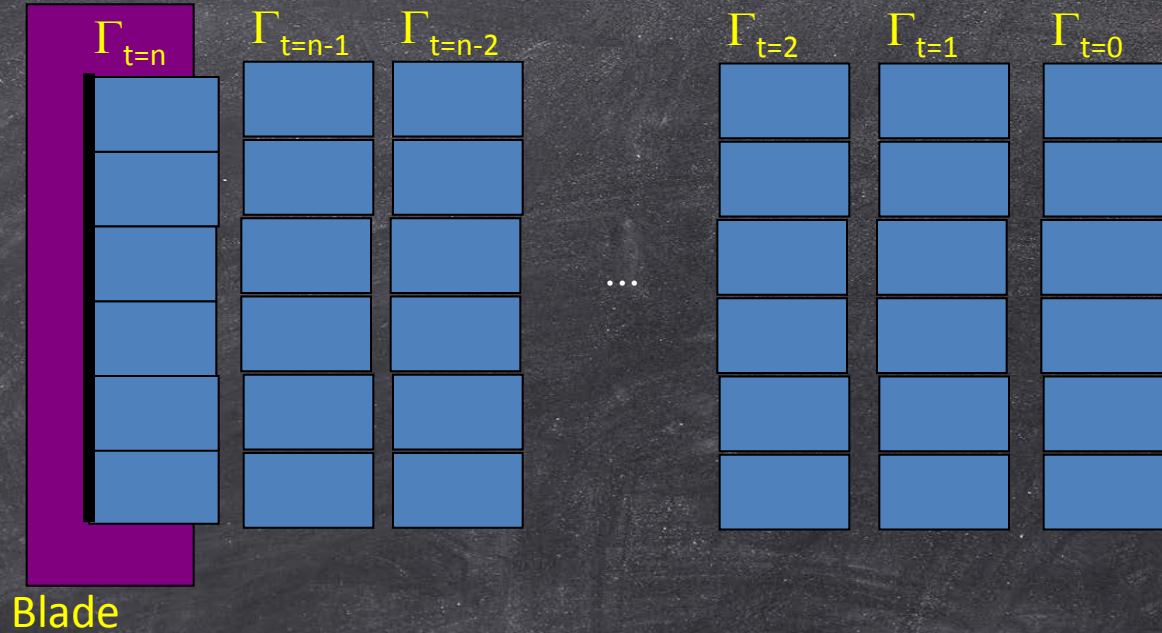
Vortex wake: ring discretization



Vortex wake: initial condition



Vortex wake: time development



Frozen vs Free Wake Vortex Model

Frozen Wake Vortex Model-

It is assumed that the geometric constraints and the unperturbed flow velocity are the only parameters affecting the wake. Wake velocity self-induction or velocities induced by the blades are neglected. No deformation of the wake occurs.

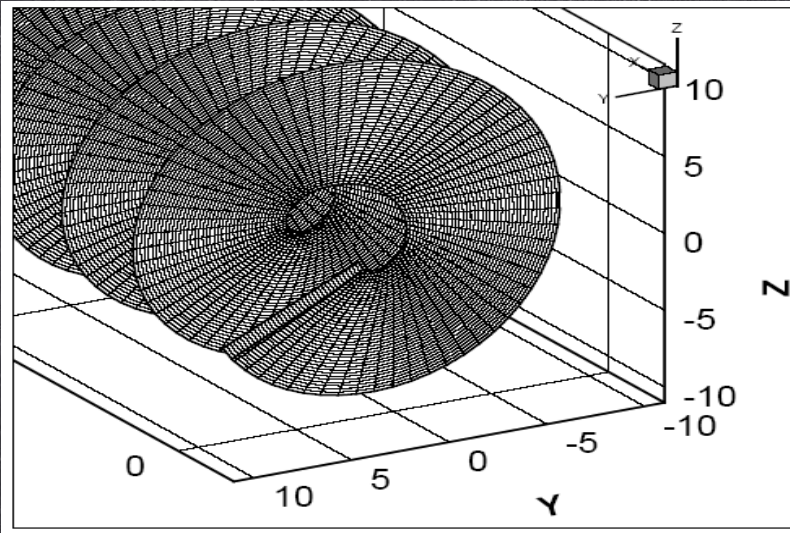
Free Wake Vortex Model-

The model takes into consideration the velocities induced on the wake by the blades' circulation and by the wake itself. Wake deformation occurs.

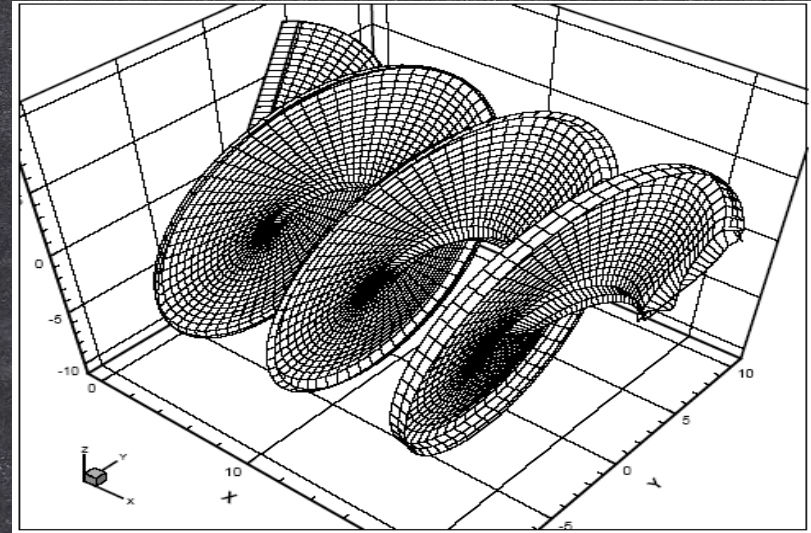


Frozen vs Free Wake Vortex Model

Frozen wake

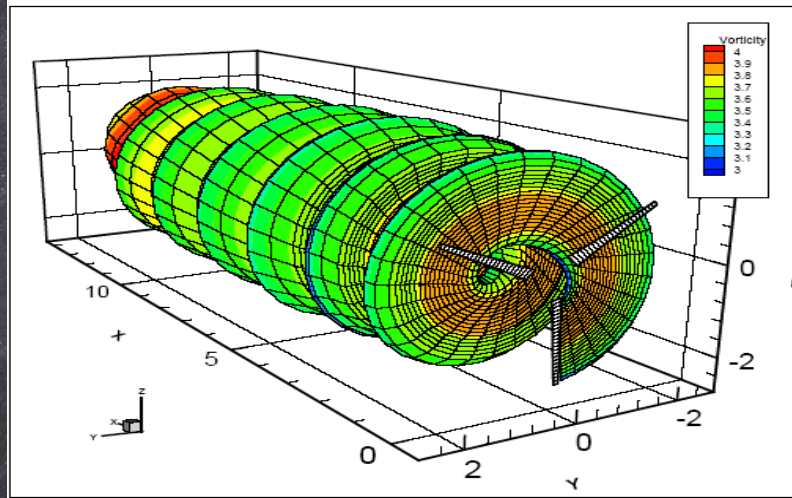


Free wake

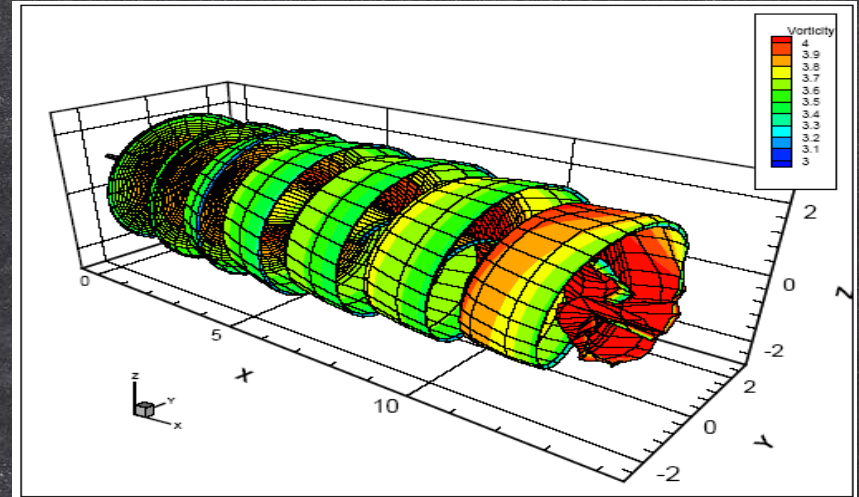


Frozen vs Free Wake Vortex Model

Front view



Back view



Free wake models consume much more time than frozen wake models

Frozen wake

For $i = 1$ to N

$$\vec{x}_i = \vec{x}_i + \vec{q}_\infty$$

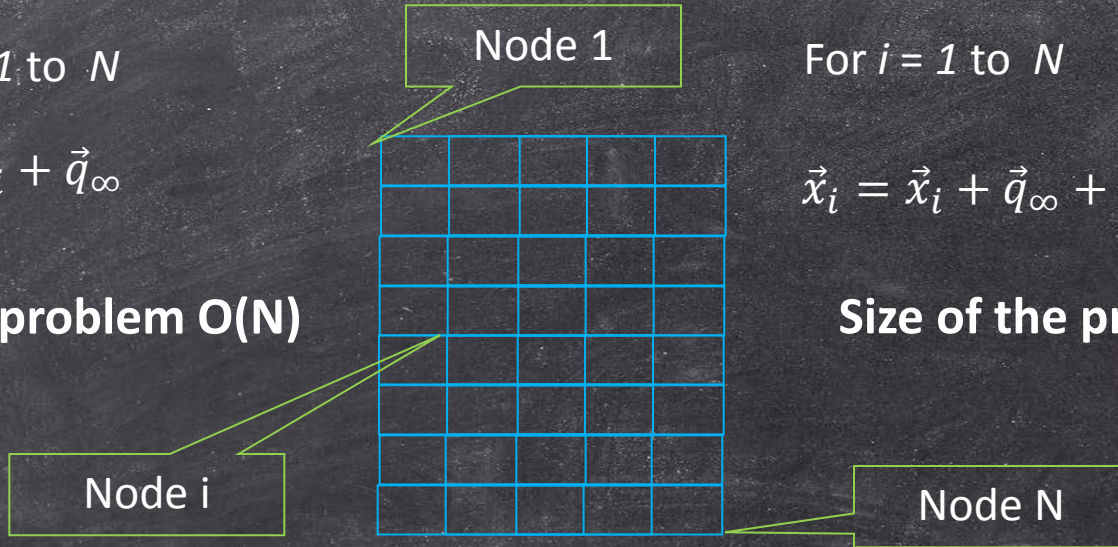
Size of the problem $O(N)$

Free wake

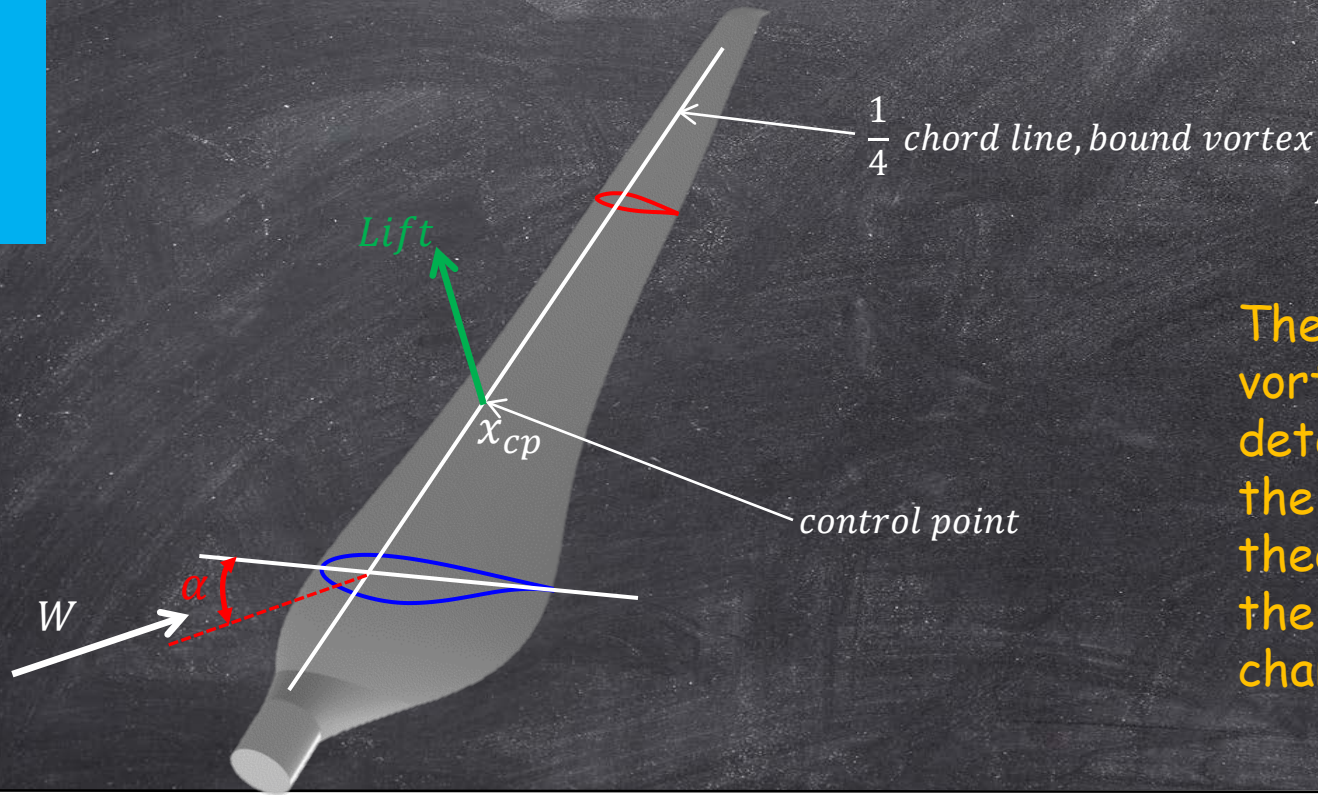
For $i = 1$ to N

$$\vec{x}_i = \vec{x}_i + \vec{q}_\infty + \sum_{j=1}^N \overrightarrow{Ind}_j(\vec{x}_i)$$

Size of the problem $O(N^2)$



The strength of the vortex line is determined locally

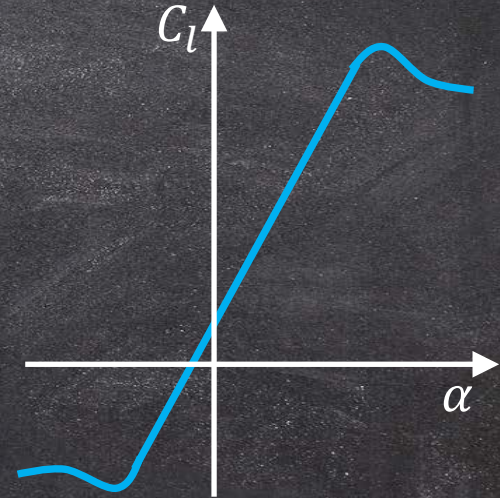
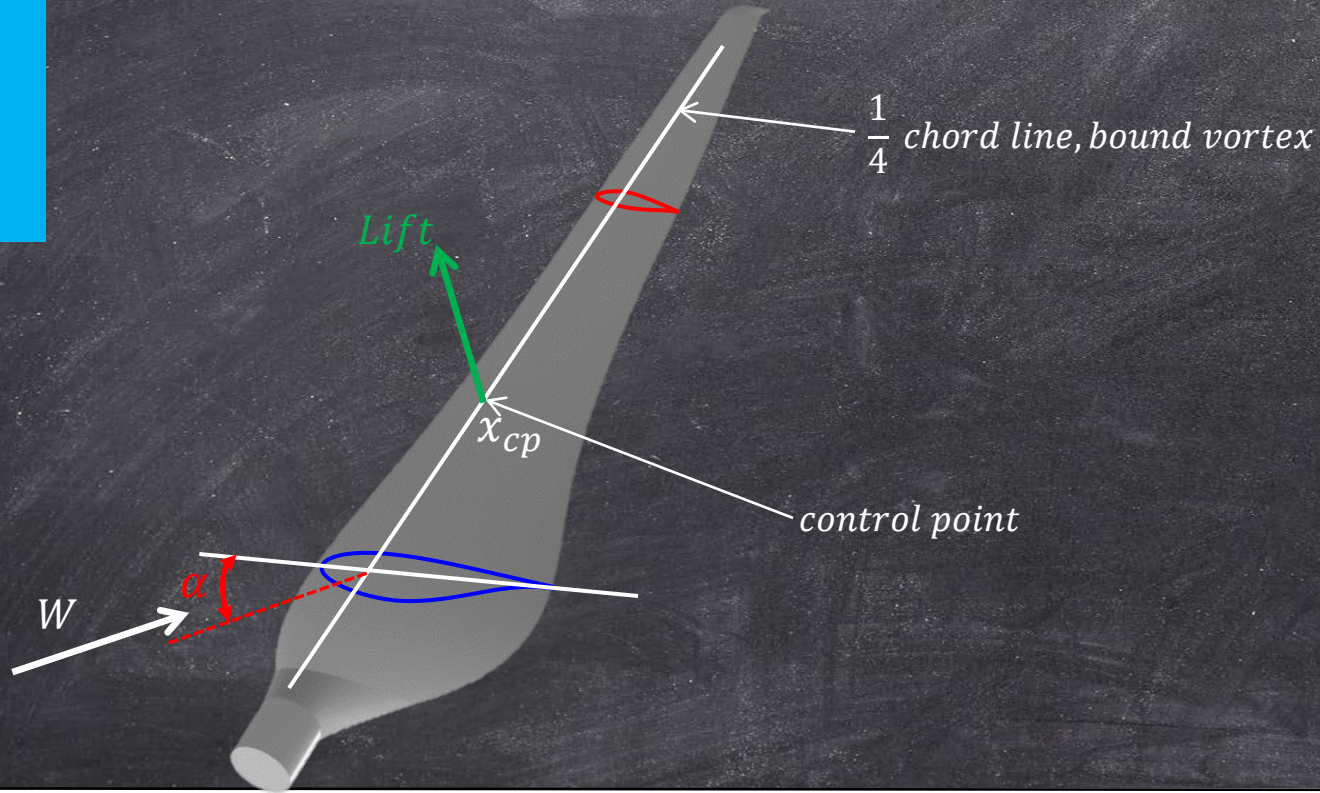


$$L = -\rho W \Gamma$$

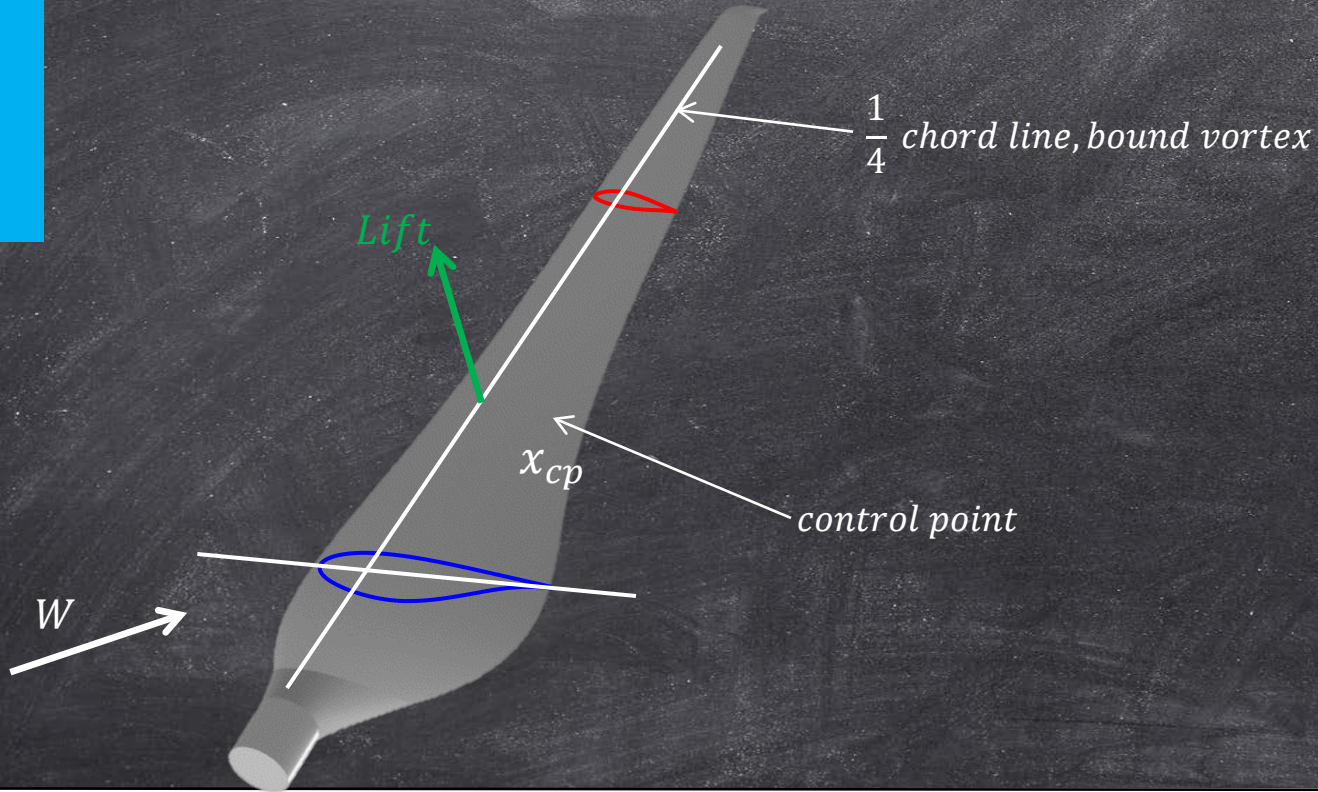
The strength of the vortex elements are determined by employing the Kutta-Joukowski theorem on the basis of the local aerofoil characteristics



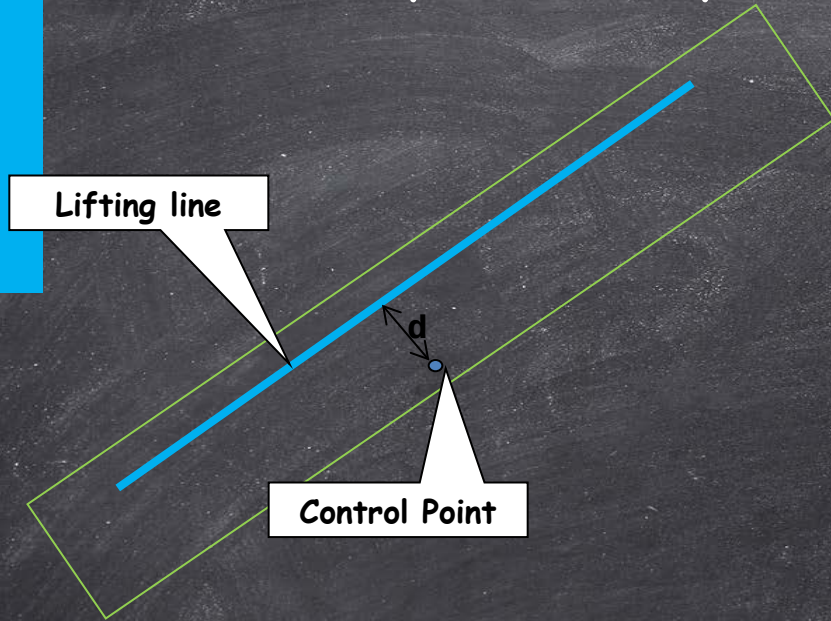
Using a angle of attack condition



Non-entry boundary condition to determine lift (I)



Non-entry boundary condition to determine lift (II)



Velocity at control point is zero
(boundary condition)

$$\vec{U}_{vor} + U_{\infty} \sin(\alpha) = 0$$

$$\vec{U}_{vor} = \frac{\Gamma}{2\pi d}$$

$$\Gamma = \frac{1}{2} U_{\infty} c C_L$$

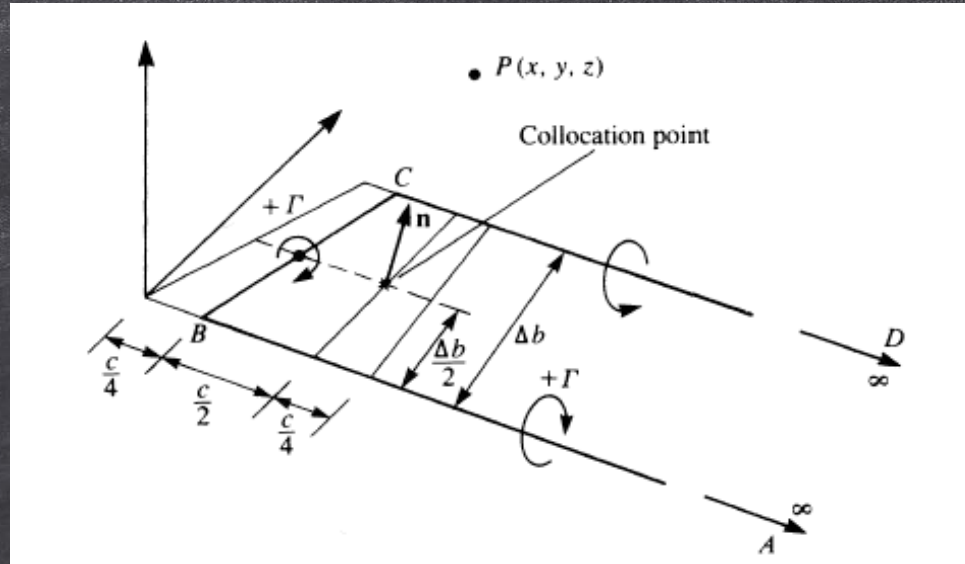
$$C_L = 2\pi \sin(\alpha)$$

Pistolessi theorem - $\frac{3}{4}$ chord

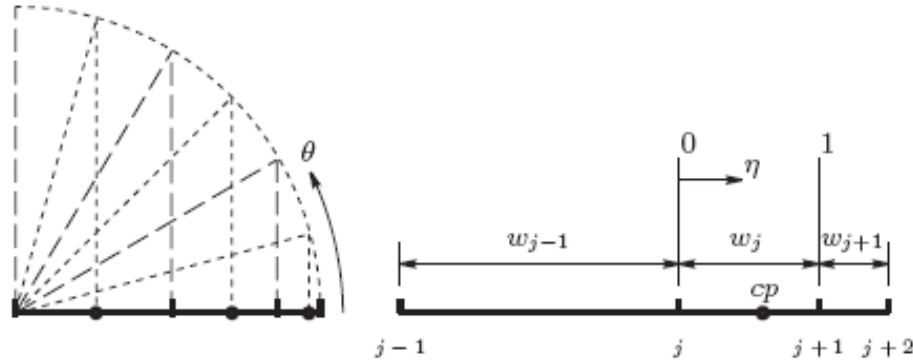
$$\Gamma = \pi U_{\infty} c \frac{\Gamma}{2\pi d U_{\infty}} \Rightarrow d = \frac{c}{2}$$



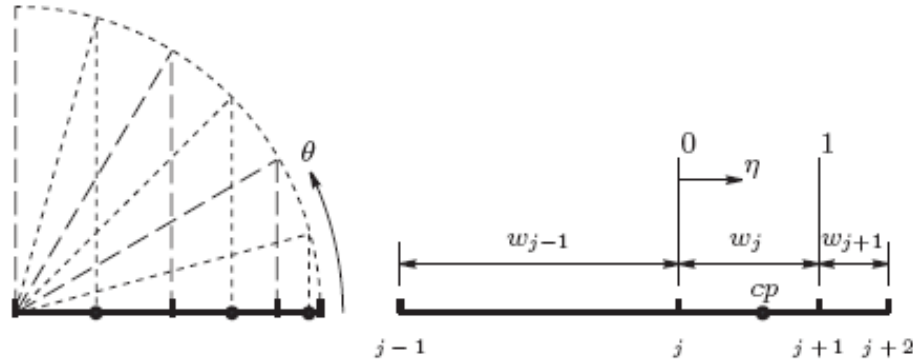
Non-entry boundary condition to determine lift (III)



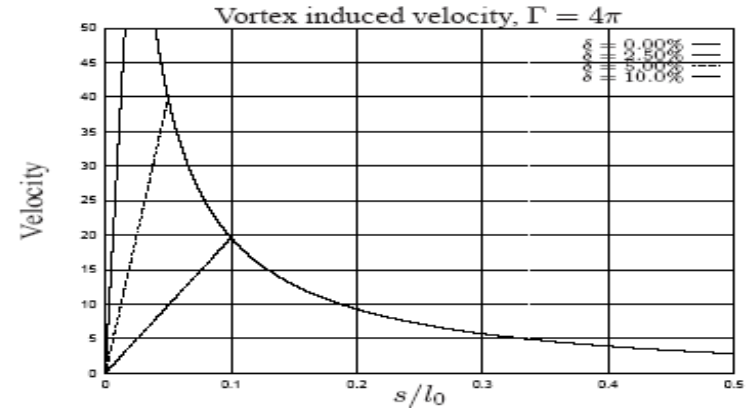
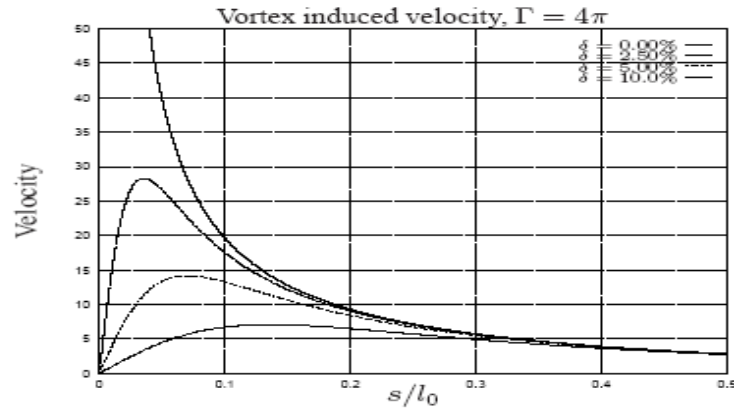
Control Points Distribution



Control Points Distribution



Modeling the Vortex Core



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